

Navigating The Future of Online Shopping

Ecommerce Analysis Using Python and SQL

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About Company

This dataset captures transactional data from a multi-category e-commerce platform that connects customers and sellers across diverse regions. The platform supports a broad product catalog and flexible payment options, including installment plans.

As competition in the e-commerce landscape intensifies, analyzing customer behavior, sales performance, and seller contributions becomes essential for driving sustainable business growth.

Tools And Technologies

- SQL (MySQL) – Data Extraction, Querying
- Python (Pandas, Matplotlib) – Data Analysis, Visualization
- VsCode – Implementation
- Workflow – Data Cleaning > Analysis > Visualization > Insights

Dataset Overview

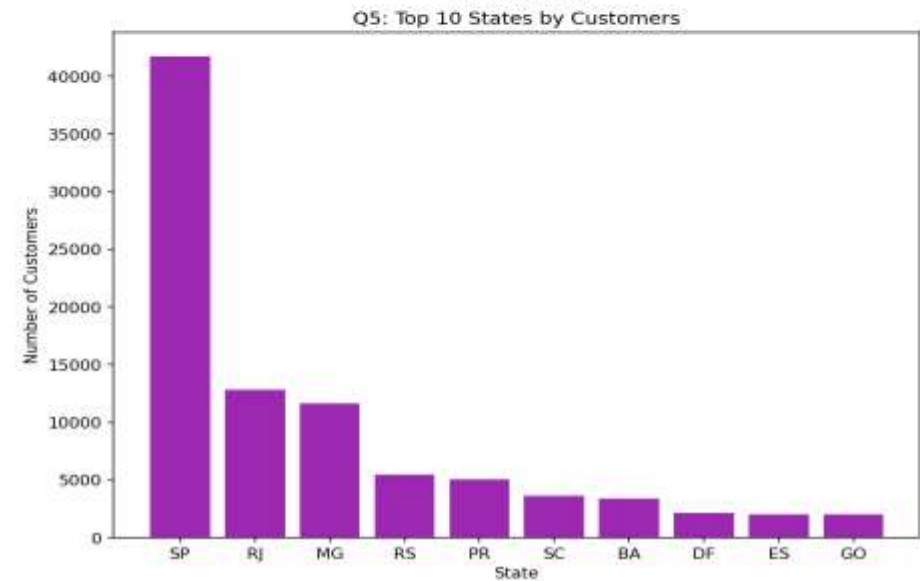
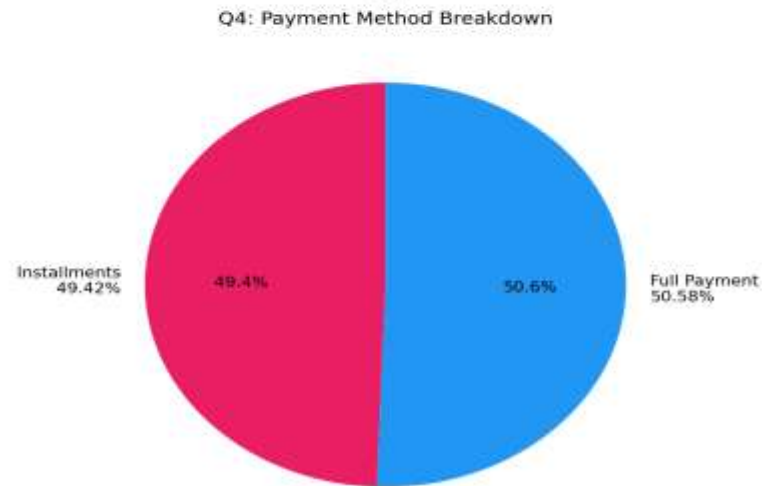
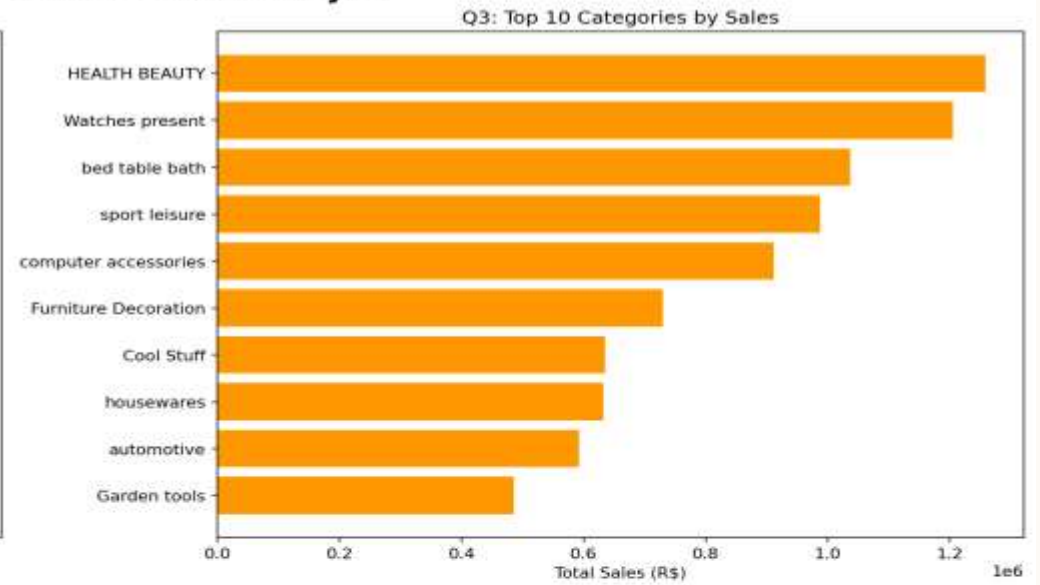
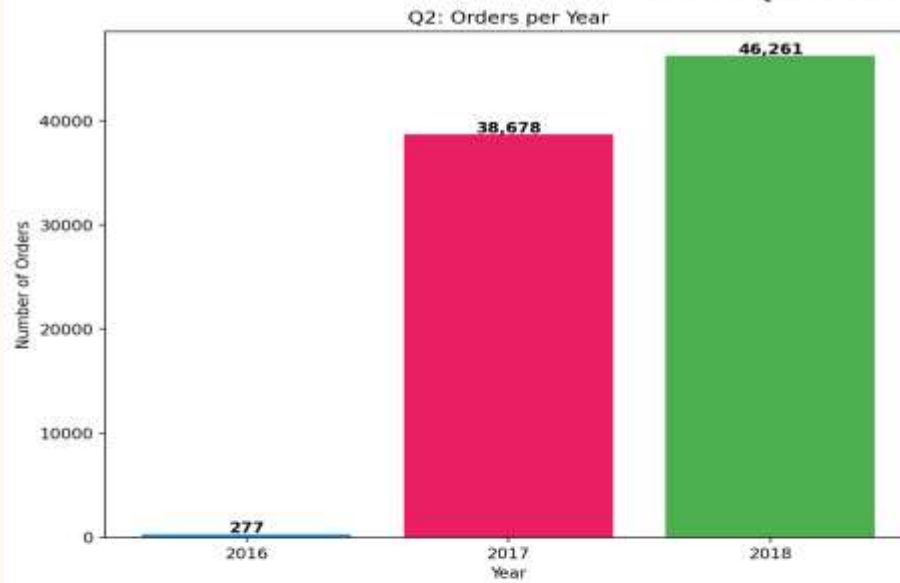
File	Description	Rows
customers.csv	Customer demographic details	99,441
sellers.csv	Seller information	3,095
orders.csv	Order history and details	99,441
order_items.csv	Order items details	112,650
payments.csv	Payment details	103,886
products.csv	Product details	32,951
geolocation.csv	Geolocation details	1,000,163

Insights - Basics

Objective: Extract fundamental insights from the dataset.

- Unique cities where customers are located – 4119 cities
- Number of orders placed in 2017 – 45,101 orders
- Total sales per category – health & Beauty Rs 1.25m
- Percentage of orders paid in installments – 49.42%
- Number of customers from each state – SP leads with 41,746

Basic Queries — E-Commerce Analysis

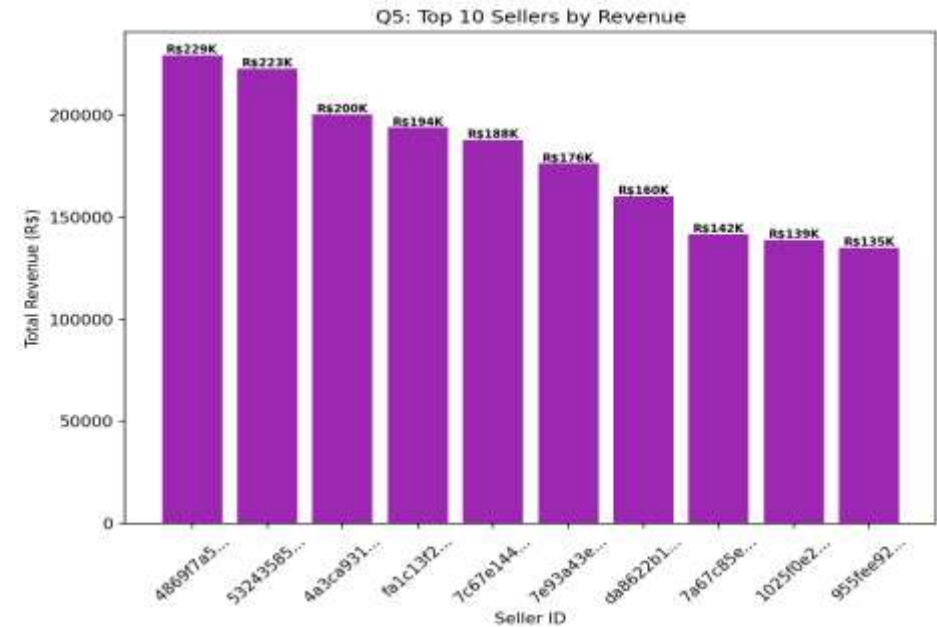
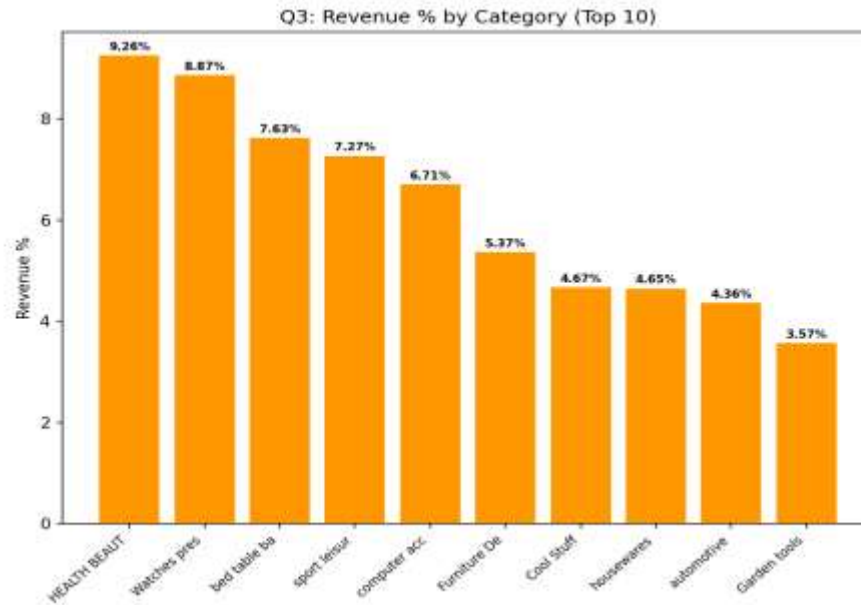
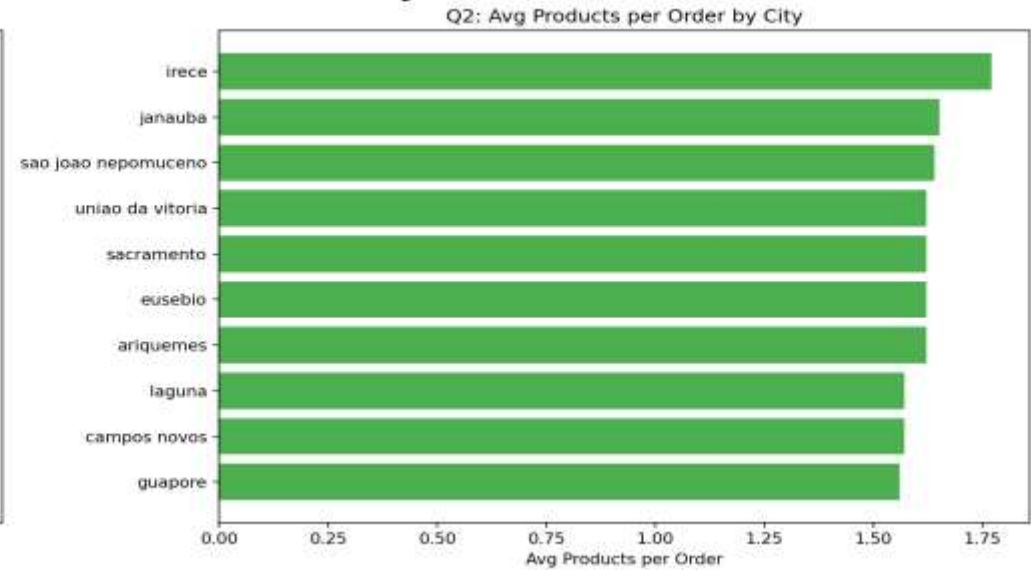
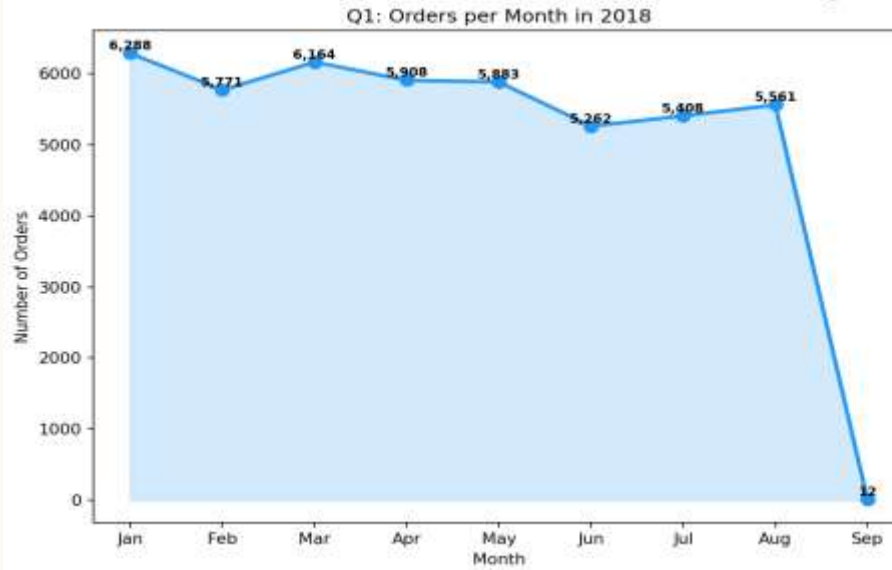


Insights - Intermediate

Objective: Dive deeper into sales and order trends.

- Orders per month in 2018 – January peak 6,288 orders
- Avg products per order by city – Padre Carvalho, 7 products/orders
- Revenue % by product category – Health & Beauty 9.26%
- Price vs purchase frequency – Most purchased products, 527 times
- Total revenue by seller ranked – Top seller, Rs 2,29,472

Intermediate Queries — E-Commerce Analysis



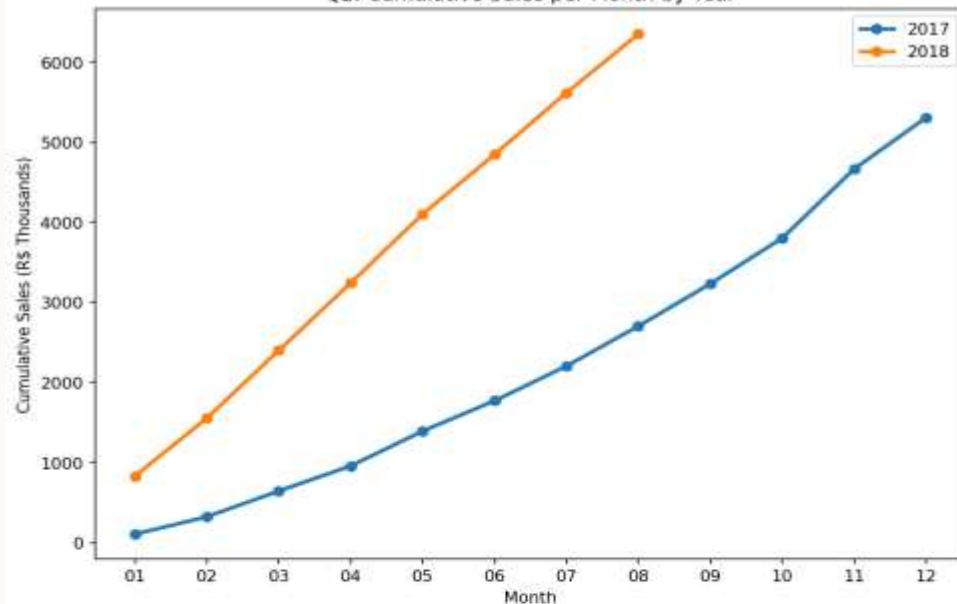
Insights - Advanced

Objective: Generate strategic and customer-centric insights.

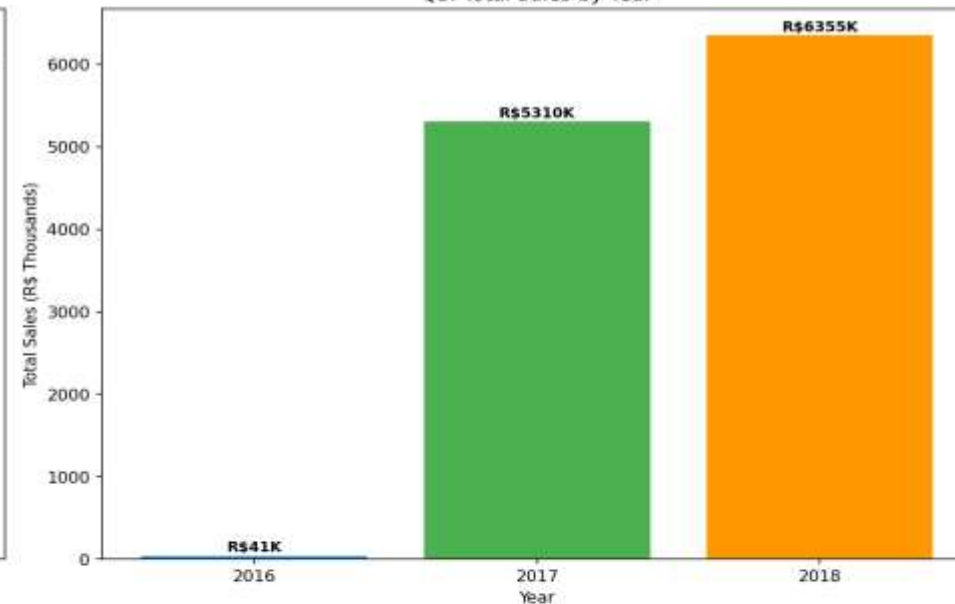
- Moving average of order values – Tracks customer spending trends
- Cumulative sales per month – 2018 hit, Rs 6.35 by August
- Year-over-year growth rate – 2017 grew 12,745% vs 2016
- Customer retention rate – 0% most buy only once
- Top 3 customers per year – 2017 top spender, Rs 13,440

Advanced Queries — E-Commerce Analysis

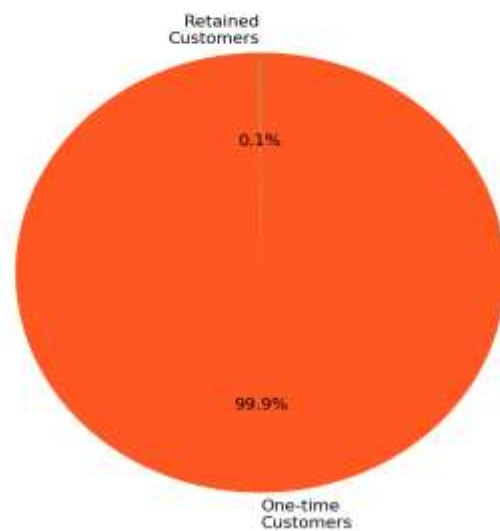
Q2: Cumulative Sales per Month by Year



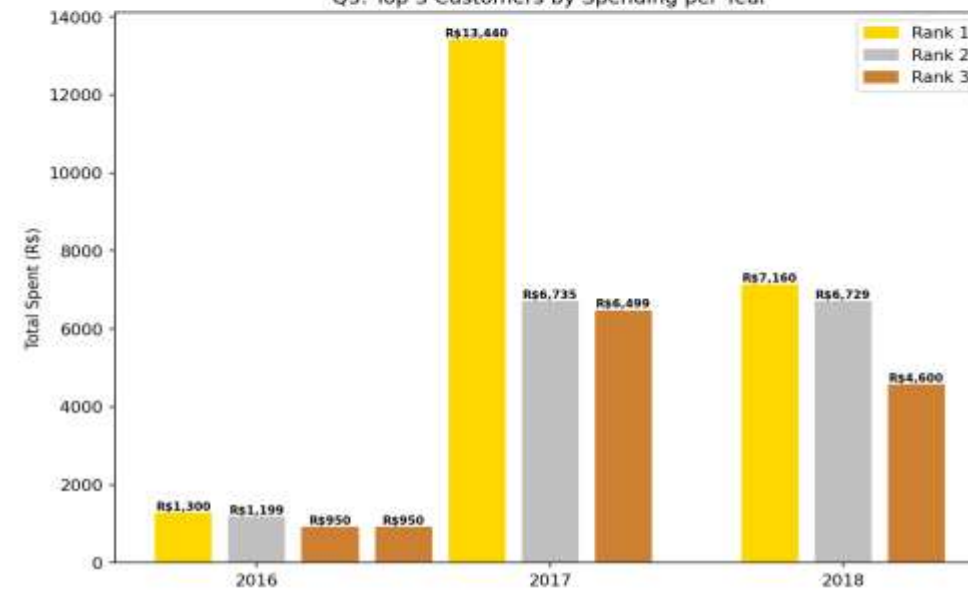
Q3: Total Sales by Year



Q4: Customer Retention Rate (6 months)



Q5: Top 3 Customers by Spending per Year



Key Business Insight

- 1. São Paulo dominates — SP state has 42% of all customers
- 2. Health & Beauty is top category — Rs1.25M in total sales
- 3. Half of customers pay in installments — 49.42% use installment plans
- 4. Massive growth in 2017 — Business grew 12,745% compared to 2016
- 5. Low retention rate — Most customers purchase only once, suggesting need for loyalty programs
- 6. Seasonal peaks — November 2017 had highest monthly sales (Rs863K) likely due to Black Friday



Thank You